

Tracking by Predicting 3-D Gaussians Over Time

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Abstract

We propose Video Gaussian Masked Autoencoders (Video-GMAE), a self-supervised approach for representation learning that encodes a sequence of images into a set of Gaussian splats moving over time. Representing a video as a set of Gaussians enforces a reasonable inductive bias: that 2-D videos are often consistent projections of a dynamic 3-D scene. We find that tracking emerges when pretraining a network with this architecture. Mapping the trajectory of the learnt Gaussians onto the image plane gives zeroshot tracking performance comparable to state-of-the-art. With small-scale fine-tuning, our models achieve 34.6% improvement on Kinetics, and 13.1% on Kubric datasets, surpassing existing self-supervised video approaches.

1. Introduction

Each pixel’s journey through a video tells a story about motion. By tracking such pixels, we can understand the structure of the scene and interaction among its constituents. In computer vision systems, solving correspondence is essential for computational photography, 3-D understanding, and other long-range reasoning tasks. While pixel tracking has been studied in depth in vision literature [10, 23, 29, 40], it has been limited to training models on annotated point tracks, bounding boxes, segmentation masks [8, 23, 40], synthetic datasets [45] or specialized architectures [21, 42, 43]. In this paper, we present a self-supervised method on videos that learns strong representations for correspondence.

We find that the representations learned by existing video self-supervised learning (SSL) approaches do not perform well on point tracking. We hypothesize that the classic (space-)time patch prediction objective does not strongly enforce temporal consistency; that is, this objective can be optimized without understanding pixel-level correspondence across a long sequence of frames.

How can we set up our SSL task such that this correspondence emerges in the representations? We note that the motion of objects in 3-D manifests as point tracking on the image plane. In this paper, we leverage this insight to learn point tracking by pretraining on unlabeled videos.

We train a Masked Autoencoder [19]-style encoder-decoder architecture that takes video as input and predicts *Gaussian primitives* [26] that move in time. We call our approach *Video-GMAE*, in which for the first frame, we predict a fixed number of Gaussian primitives. For subsequent frames, we predict the *translation* and *color* change of each Gaussian with respect to the previous frame. In this way, each Gaussian preserves its identity over the span of N frames. Encoding a frame sequence as a set of moving Gaussians explicitly imposes temporal correspondence in 3-D as an inductive bias in our training. This inductive bias makes the SSL task *harder*, inducing the latent representations to encode long-term correspondence.

To investigate our pretraining methodology, we devise an algorithm to compute point tracks from the predicted 4-D Gaussians *zeroshot*. We find that a zeroshot tracker naturally emerges from our correspondence-aware Gaussian representations. *Video-GMAE* also gives latent representations that are useful for tracking. Fine-tuning *Video-GMAE* on tracking datasets outperforms prior supervised and self-supervised methods. *Video-GMAE* also outperforms other video SSL approaches such as VideoMAE [48], MAE-ST [14] on frozen encoder tracking evaluation.

Broadly, we aim to bring classic end-to-end SSL together with differentiable rendering. Differentiable rendering is a natural inductive bias for video SSL: since videos are basically 2-D projections of a 3-D world, dynamic yet consistent over time. We hope that this work aids the search for video-SSL methods that learn longer-horizon and more generalizable representations.

2. Related Work

Self-supervised Learning: Over the years, self-supervised pretraining has shown strong performance across domains like language, vision, and robotics. In computer vision,

Project page: <https://videogmae.org/>. Code: <https://github.com/tekotan/video-gmae>. Corresponding author: tanish@berkeley.edu.

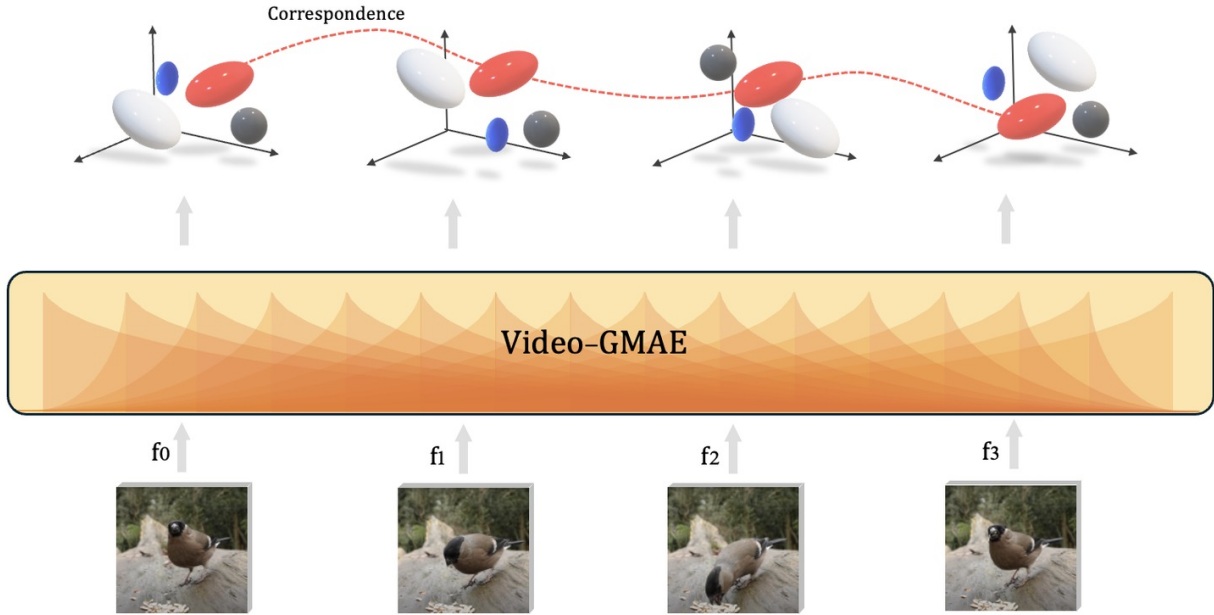


Figure 1. **Self-supervised Video Pretraining for Correspondence:** Given a sequence of video frames, our approach *Video-GMAE* predicts Gaussian primitives for each frame to reconstruct the whole video. In addition to this, we also enforce correspondence in the Gaussian primitives by predicting the delta Gaussians for all but the first frame.

there are broadly two schools of thought: discriminative and reconstructive pretraining. Discriminative methods typically train a model to recognize that different augmentations of the same image should be close in feature space. Early works like [52] and SimCLR [6] demonstrated that contrastive learning over such instance discrimination tasks can yield strong visual features. Later methods like MoCo [18] and DINO [4] further pushed this line of work, showing that such learned representations can transfer well to a range of downstream tasks.

Reconstructive pretraining learns to model the data distribution by trying to reconstruct an image or a video from its noisy version. One of the most successful methods for such pretraining in computer vision has been the BERT [9]-style masked modeling of images proposed by BEiT [1], and MAE [19]. Compared to BERT, MAE uses asymmetric encoder-decoders, allowing it to be very efficient at training with high masking ratios. This style of reconstructive pretraining learns strong visual priors and shows impressive results on various downstream tasks such as object detection [31], pose estimation [53], and robot tasks [37]. Extending this to videos, VideoMAE [50] and MAE-ST [14] showed that, with large masking ratios, MAEs can learn very strong representations from unlabeled videos. Another line of generative video modeling uses autoregressive models to simply predict the next patch or next frame [39, 41]. Recently, these models have been used as an encoder for vision-language models [15, 47].

Gaussian Splatting: Gaussian Splatting [26] is a recent differentiable rendering method that uses 3-D Gaussian primitives as the underlying representation, allowing for flexible optimization and high-quality reconstructions. This idea builds on a broader trend in differentiable rendering, which has become a popular way to connect 3-D geometry with 2-D image supervision. By making the rendering process differentiable, these methods enable gradient-based learning of 3-D structures, like meshes and point clouds, from images. For instance, [33] introduced a soft rasterizer for mesh-based rendering, while [28] proposed an efficient differentiable renderer for large point clouds. NeRF [35] and Mip-NeRF [2] extend this idea to volumetric scene representations, using differentiable volume rendering [30] to learn 3-D radiance fields from just a few multi-view images.

Point Tracking: Tracking has been studied in computer vision under different scales. Traditional methods like the Kanade–Lucas–Tomasi tracker [46] have been widely used for this purpose, leveraging local image gradients to track features over time. Recent deep-learning driven advancements have introduced more robust and flexible approaches. For instance, RAFT [45] extracts per-pixel features per-frame and uses correlation across frames to compute point tracks. The Tracking Any Point (TAP) [10] paradigm focuses on tracking arbitrary points on deformable surfaces, accommodating complex motions and occlusions. Models like TAPIR [11] enhance this capability by employing per-frame initialization and temporal refinement strategies, enabling

accurate tracking of points across diverse scenarios. Both the above methods are trained using supervised learning with synthetic data. Another line of work focuses on developing SSL algorithms for point tracking. CRW [51] models the evolution of patches over time as a random walk parameterized by a learnable matrix, trained using cycle consistency. GMRW [42] further extends it to pixel-level tracking. DIFT [44] shows that nearest neighbour on patch-level features extracted from pretrained diffusion models provides not just temporal, but also semantic correspondence across frames.

Benchmark datasets like TAP-Vid [10] and TAPVid-3-D [27] have been developed to evaluate and compare the performance of various point tracking methods, providing standardized metrics and diverse testing scenarios. Overall, the evolution of point tracking techniques, from classical methods to modern deep learning-based approaches, reflects the ongoing efforts to achieve more accurate, robust, and versatile tracking capabilities in computer vision applications.

3. Preliminaries

3.1. Self-supervised Masked Autoencoders

Masked autoencoders learn data representations by randomly masking parts of the input and training the model to predict the missing content. In language, BERT [9] follows this approach by using a transformer [49] to predict masked text tokens. For images, methods like MAE [19] and BEiT [1] train the model to reconstruct masked image patches. In videos, approaches such as VideoMAE [48] and MAE-ST [14] extend this idea by masking spatio-temporal patches across frames. MAE-ST uses a Vision Transformer (ViT) [13] encoder to process the visible patches, and a lightweight ViT decoder that ingests visible and masked tokens to reconstruct the missing video content.

3.2. 3-D Gaussian Splatting

3-D Gaussian Splatting originally introduced for optimization-based single-scene 3-D reconstruction [26] was later extended to image-level representation learning in [38]. Each primitive is defined by a 3-D center position $p \in \mathbb{R}^3$, a covariance matrix $\Sigma \in \mathbb{R}^{3 \times 3}$, a color $r \in \mathbb{R}^3$, and an opacity value $o \in \mathbb{R}$, collectively encoding the geometry and appearance of the scene. During rendering, the Gaussians are transformed into the camera frame and projected onto the image plane using volumetric splatting. This rendering pipeline is fully differentiable, allowing gradients to flow back to the Gaussian parameters from the rendered output. Following standard practice, the covariance matrix is factorized as $\Sigma = RSS^T R^T$, where $S = \text{diag}(s) \in \mathbb{R}^{3 \times 3}$ is a diagonal scaling matrix parameterized by $s \in \mathbb{R}^3$, and $R \in \mathbb{R}^{3 \times 3}$ is a rotation matrix represented via a quaternion $\phi \in \mathbb{R}^4$. As a result, each Gaussian is fully described by a 14-dimensional vector $g = \{\mu, s, \phi, r, o\} \in \mathbb{R}^{14}$.

4. Self-Supervised Pretraining

Our self-supervised pretraining method is summarized in Fig 2. We feed $k = 16$ RGB frames into a ViT encoder as video patches with a masking ratio of 95% as in [14]. The encoder computes latents, which are passed into a decoder concatenated with query tokens. The decoder generates $k \times n$ Gaussian primitives for rendering. The first n of these are treated as independent Gaussian primitives in free space for the first frame. The rest are *residual* Gaussians deltas, which are integrated to compute Gaussians for subsequent frames. Predicting residuals for the first set of Gaussians enforces correspondence in the Gaussian primitives over time.

We start with an image-based encoder trained to predict 3-D Gaussians for an image [38]. We extend this model to be a video encoder by adding a separate, learnable positional embedding for the time axis. We train this video model on k frames, with 16×16 patches per frame. We do patchification at the frame level to utilize the image model–GMAE [38].

The decoder takes in encoder latents and $k \times n$ learnable query tokens, $k = 16$ frames at pretraining, and $n = 256$ Gaussians per frame. Define $G_0 = \{g_1, g_2, \dots, g_n\}$ as the set of Gaussians for the first frame. For subsequent frames, the decoder only predicts delta Gaussians, $\Delta G_1, \dots, \Delta G_{k-1}$. We find that predicting $\Delta G = \{\{\Delta\mu_1, \Delta r_1\}, \dots, \{\Delta\mu_n, \Delta r_n\}\}$ is sufficient. During pretraining, we integrate the delta Gaussians per frame $G_t = \Delta G_t + G_{t-1}$ to get the k sets of Gaussians primitives that we render using Gaussian splatting, [26] and train the model end-to-end with reconstruction loss. This formulation allows us to train the model with correspondence under self-supervised training.

Our pretraining dataset combines the training sets of Kinetics[25], Kubric [16] and Davis [36]. *We only use the unlabeled videos from these datasets.* More details about pretraining methodology can be found in the supplementary.

5. Zeroshot Point Tracking

In this section, we present an algorithm to extract point tracks from the collection of Gaussian primitive trajectories predicted by *Video-GMAE*. At a high level, we splat the Gaussian primitives on the image plane with the RGB values of each Gaussian replaced with the projected 2-D $\Delta\mu_i$. This renders the 2-D Gaussian mean deltas into a flow field. Following the flow for any point tracks it to the next frame. A compact pseudocode summary is provided in the supplementary (Algorithm 1).

Given an initial set of Gaussians $G_0 = \{g_1, g_2, \dots, g_n\}$ for frame $t = 0$, each primitive $g_i = \{\mu_i^{(0)}, s_i, \phi_i, r_i^{(0)}, o_i\}$ contains a 3-D position $\mu_i \in \mathbb{R}^3$. For subsequent frames, the model predicts residual updates $\Delta G_t = \{\Delta\mu_i^{(t)}, \Delta r_i^{(t)}\}_{i=1}^n$ to the Gaussian means. We apply these deltas recursively to

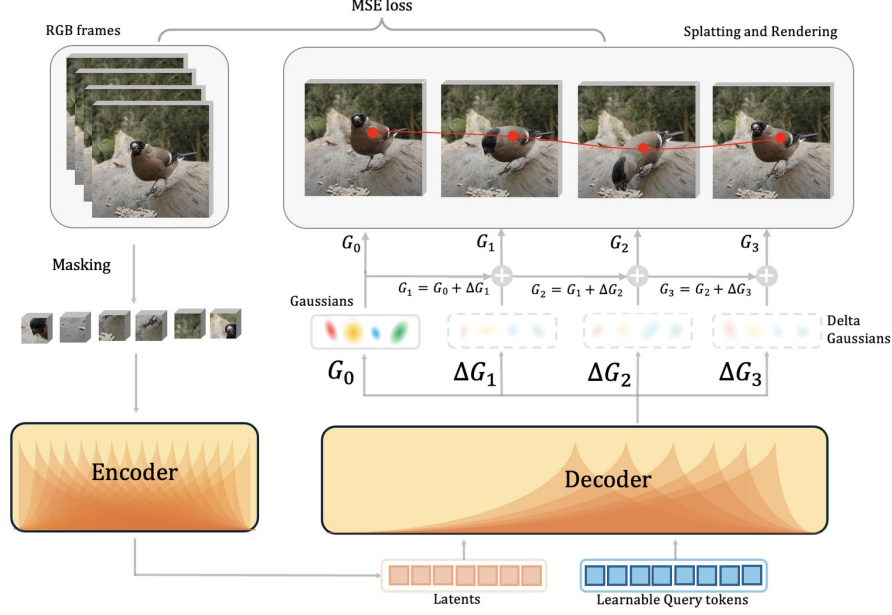


Figure 2. **Video Masked Autoencoding via Gaussian Splatting:** The ViT Encoder processes masked input frames to produce latent embeddings. The ViT Decoder then predicts explicit Gaussian parameters for frame f_1 based on query tokens, including color, opacity, center, scale, and orientation, and the Gaussian deltas for the rest of the frames. The explicit Gaussians for frame f_1 to f_t are calculated and rendered via differentiable volume splatting to reconstruct all the frames. We pre-train our models fully end-to-end with self-supervision.

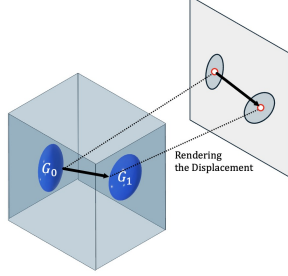


Figure 3. **Zeroshot Point Tracking:** The 3-D centers of the predicted Gaussian primitives are projected and the subsequent 2-D displacement vector is rendered.

get the means of the Gaussian primitives as:

$$\mu_i^{(t+1)} = \mu_i^{(t)} + \Delta\mu_i^{(t)}. \quad (1)$$

To extract motion in the image plane, we project the means of each Gaussian into pixel coordinates using the same camera intrinsics $K \in \mathbb{R}^{3 \times 3}$ and extrinsics $[R | t] \in \text{SE}(3)$, that we use to render during the training phase. Let $\Pi(\cdot)$ denote the perspective projection:

$$x_i^{(t)} = \Pi \left(K[R | t], \mu_i^{(t)} \right). \quad (2)$$

We also use the projected Gaussian locations to define a per-Gaussian image-plane offset:

$$\Delta x_i^{(t+1)} = x_i^{(t+1)} - x_i^{(t)}. \quad (3)$$

We calculate the instantaneous 2-D displacement vector $\Delta x_i^{(t)}$ carried by each Gaussian, encode them as pseudo-RGB values $c_i^{(t)} = (\Delta x_{i,x}^{(t)}, \Delta x_{i,y}^{(t)}, 0)$, and splat them onto the image plane using standard volumetric alpha compositing. The resulting dense flow field $F^{(t)} \in \mathbb{R}^{H \times W \times 2}$ at each pixel $u \in \mathbb{R}^2$ is computed by an opacity-weighted average of all displacements:

$$F^{(t)}(u) = \sum_{i=1}^n \alpha_i^{(t)}(u) \cdot \Delta x_i^{(t)}, \quad (4)$$

where $\alpha_i^{(t)}(u) \in [0, 1]$ is the rasterized visibility of Gaussian i in the pixel u , calculated using differentiable Gaussian splatting [26]. We extend this notation for an arbitrary point p using bilinear interpolation, where $\alpha_i^{(t)}(p) = \text{bilinear}(\alpha_i^{(t)}, p)$ and $F^{(t)}(p) = \text{bilinear}(F^{(t)}, p)$.

To improve robustness around occlusions, we additionally leverage the renderer’s soft per-pixel assignments. First, we fix a per-point anchor set $\mathcal{S} = \text{Top}_k(\{\alpha_j^{(t)}(u)\}_{j=1}^n)$ chosen once at $t = 0$. Then, we define the per-frame anchor mass:

$$\omega^{(t)} = \sum_{i \in \mathcal{S}} \alpha_i^{(t)}(p^{(t)}), \quad (5)$$

and re-normalized mixture weights over this fixed set:

$$\tilde{\pi}_i^{(t)} = \frac{\alpha_i^{(t)}(p^{(t)})}{\sum_{j \in \mathcal{S}} \alpha_j^{(t)}(p^{(t)}) + \varepsilon}, \quad i \in \mathcal{S}. \quad (6)$$

The point is visible if $\omega^{(t)} \geq \tau_{\text{vis}}$ and occluded otherwise.

We write $a^{(t)} := p^{(t)} + F^{(t)}(p^{(t)})$ for $p^{(t)}$ advected by $F^{(t)}$ and $s^{(t+1)} := \sum_{i \in \mathcal{S}} \tilde{\pi}_i^{(t)} (x_i^{(t)} + \Delta x_i^{(t+1)})$ for the weight-averaged anchor Gaussian primitives plus their displacements. Then, the per-frame update for $p^{(t)}$ is:

$$p^{(t+1)} = \begin{cases} (1-\beta)a^{(t)} + \beta s^{(t+1)} & \omega^{(t)} \geq \tau_{\text{vis}} \\ s^{(t+1)} & \omega^{(t)} < \tau_{\text{vis}}. \end{cases} \quad (7)$$

Here $\beta \in [0, 1]$ blends pure flow-based advection with the top- k Gaussian-mixture proposal. Setting $k = n$ and $\tau_{\text{vis}} = 0$ recovers a pure flow-based update that ignores occlusions. If the updated location $p^{(t+1)}$ lies outside the image dimensions, we mark the point as occluded. The hyperparameters $k = 8$, $\tau_{\text{vis}} = 0.5$, and $\beta = 0.3$ are determined based on performance on the Kubric train dataset.

6. Supervised Fine-tuning for Point Tracking

We evaluate the encoder’s learned representation on point tracking to assess the improved correspondence-based features. For point tracking evaluation, we adopt a cross-attention-based readout architecture and evaluate it on the TAP-Vid protocol, which includes three datasets: TAP-Vid-Kinetics[10], TAP-Vid-Davis[10], and Kubric[16]. Each evaluation run uses a single query per point track. We follow the evaluation protocol described in [42]. We either freeze (*Video-GMAE-frozen*) or fine-tune (*Video-GMAE-finetune*) the encoder, and the readout network consumes the spatio-temporal encoder features to predict tracked points across frames, as shown in Fig 4.

Following the design in [5], we first apply layer normalization to the encoder features, add learnable temporal embeddings, and use 64-dimensional Fourier-based positional queries per frame, projected to the token feature dimension. These queries cross-attend to the processed encoder features using a 16-headed attention mechanism. This is followed by a residual MLP with hidden size four times the token feature dimension and GeLU [20] activation, and finally a linear layer mapping with sigmoid activation to a 3-D vector containing the tracked 2-D point and its occlusion.

For frozen evaluation, we train only the readout layers. We train each model on one A100, with batch size of 8, and 16 tracks per video for 50k steps, using the AdamW optimizer (weight decay 5e-2) and learning rate sweep over {5e-5, 1e-4, 3e-4}, selecting the best-performing configuration. In the finetuned setting, the same setup is used, except both the encoder and the readout are trained end-to-end.

7. Experiments

Evaluation: For point tracking evaluation, we report three metrics that collectively measure accuracy, robustness, and occlusion handling. Average Jaccard Score (AJ) computes

the intersection-over-union between predicted and ground-truth visible regions, reflecting alignment quality over time. Average Points within Threshold (δ_{avg}^x) reports the percentage of predictions falling within a small pixel radius of the ground truth, offering an interpretable precision metric. Occlusion Accuracy (OA) measures the frequency with which the model accurately predicts whether a point is visible or occluded in each frame. We conduct a series of experiments to evaluate the effectiveness and design choices of our proposed model. Unless otherwise specified, all experiments are performed with a frozen encoder, focusing exclusively on optimizing the lightweight decoder.

Implementation Details: We pretrained the models on 64 V100s for 90 epochs with a batch size of 128, a learning rate of 1e-3, using the AdamW [34] optimizer (weight decay 5e-2). We used 2000 warm-up steps and cosine decay for the learning rate, and gradient clipping with a value of 2.0.

7.1. Zeroshot Tracking Results

We compute AJ, δ_{avg}^x , and OA for zeroshot tracking on all three datasets following [10]. Table 2 shows the zeroshot tracking metrics. Fig 5 shows qualitative examples of the zeroshot tracking. Our zeroshot tracking results on Kubric and TAP-Vid Davis are comparable to GMRW-C [42] numbers and outperform all self-supervised methods on TAP-Vid Kinetics. The zeroshot results show the benefit of our self-supervised approach, which is comparable to other self-supervised approaches while being scalable.

To better understand the behavior of our zeroshot tracker relative to the strongest self-supervised baseline, we conduct a qualitative comparison between *Video-GMAE-zeroshot* and GMRW-C [42] on challenging sequences. Fig 6 shows hand-picked examples from TAP-Vid Kinetics that highlight both strengths and limitations of our approach. Overall, we observe that *Video-GMAE-zeroshot* produces more temporally stable tracks; once a point is visible, our method tends to persist it with minimal flickering, while GMRW-C often exhibits frame-to-frame identity switches or short-lived disappearances. *Video-GMAE-zeroshot* is also more conservative around occlusions; it predicts tracks that are typically maintained as visible until they fully leave the frame (or sometimes one to two frames thereafter), whereas GMRW-C frequently marks points as occluded 1–3 frames early and sometimes re-activates them spuriously, which degrades OA. Also, we observe that GMRW-C has a tendency to jitter points around their true locations, while *Video-GMAE* yields smoother, more conservative motion estimates. We also find that GMRW-C can be advantageous in certain regimes. When small, visually detailed regions in the video undergo large displacements, GMRW-C maintains their trajectories better than our model. Modeling each scene with only 256 Gaussian primitives constrains the spatial resolution at which fine details can be represented and tracked. This behavior is

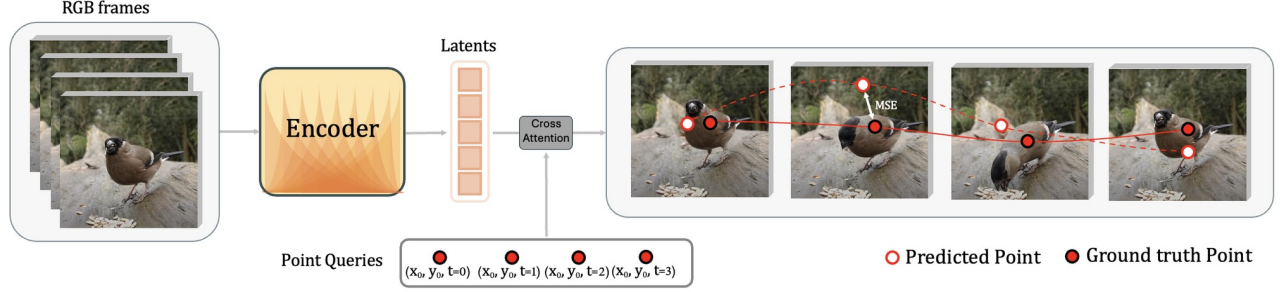


Figure 4. **Fine-tuning for Point Tracking:** We use our pretrained encoder without masking, and query the latents to predict the best point tracks. We finetune this model, using the annotated Kubric [16] dataset. The Fourier embeddings of the initial queries are calculated, and they cross-attend to the encoder latents in the fine-tuned cross-attention readout layer.



Figure 5. **Zero shot Qualitative Results:** We show qualitative results (16 frames, we only show every other frame here) from our *Video-GMAE*-zeroshot method. Without any track labels, the model can track objects with camera motion and pose changes, and shows robust tracking over long videos. A higher-resolution version can be found in the appendix (Fig. 9).

consistent with a current limitation of our Gaussian representation. These qualitative differences complement the quantitative metrics and suggest that our Gaussian-based zeroshot tracker is particularly well-suited for stable, occlusion-aware tracking, while future work could focus on improving fine-detail fidelity under large motions. More comparisons and tracking samples can be found in the supplementary.

7.2. Comparison with Video Pretraining Methods

We compare *Video-GMAE* with existing state-of-the-art self-supervised pretraining methods on the task of tracking. Specifically, we compare against VideoMAE [48] and MAE-ST [14]. Following the encoder and decoder configurations of the MAE Base and MAE Large architectures, we train two versions of our model: *Video-GMAE*-base and *Video-GMAE*-large, respectively. These are compared against VideoMAE and MAE-ST using the same frozen-encoder training setup. Table 1 contains our results. *Video-GMAE* outperforms the other pretraining baselines on all datasets.

7.3. Comparison with Tracking Baselines

We compare *Video-GMAE* with state-of-the-art self-supervised methods like DIFT [44] and GMRW [42] and supervised tracking methods like LocoTrack [7] and Co-



Figure 6. **Zero shot Comparison with GMRW-C:** We show qualitative comparisons between *Video-GMAE*-zeroshot and GMRW-C [42] on challenging sequences. *Video-GMAE* produces more temporally stable tracks, with fewer flickers and smoother motion, and tends to keep points visible until they leave the frame, while GMRW-C often occludes points 1–3 frames early. In the first example, where small, visually detailed regions move large distances across the frame, GMRW-C tracks them more accurately than our model, highlighting a limitation of our approach. A higher-resolution version can be found in the appendix (Fig. 10).

Tracker3 [24]. We show quantitative results in Table 2, and Figure 7 shows some qualitative results of the fine-tuned point tracking. To compare with other models, we use a stride of five for evaluations [10]. We train two models, *Video-GMAE*-base and *Video-GMAE*-large. For each model, we train with a frozen encoder and a fully fine-tuned encoder. *Video-GMAE*-finetune outperforms the baselines at all model scales on Kubric and TAP-Vid Kinetics benchmarks. We find that the model falls behind CoTracker [23], CoTracker3 [24], LocoTrack [7], and BootsTAPIR [12] for the TAP-Vid Davis benchmark. Among frozen encoder evaluations, we find that the large models outperform the base models. We see a similar scaling trend with the fine-tuned models, which outperform the frozen encoder models. To have a fair comparison,

Model	Kinetics			Davis			Kubric		
	AJ↑	δ_{avg}^x ↑	OA↑	AJ↑	δ_{avg}^x ↑	OA↑	AJ↑	δ_{avg}^x ↑	OA↑
MAE-ST[14]	42.3	49.8	95.4	28.3	37.2	90.1	41.5	51.6	95.9
VideoMAE[48]	46.9	54.3	94.9	31.8	39.8	90.4	44.8	55.2	95.8
<i>Video-GMAE</i>	65.1	72.0	97.4	46.7	55.7	91.8	62.4	71.9	96.6

Table 1. Cross-dataset comparison of pre-trained backbones (all are with a frozen encoder) comparing *Video-GMAE* to VideoMAE and MAE-ST. Even though we use the same masked auto-encoding, our correspondence-aware decoder forces the model to learn better representations for point tracking. We compare using a stride of five frames, the same as in Table 2.

Method	Kubric			Davis			Kinetics		
	AJ ↑	$\langle \delta_{\text{avg}}^x \rangle$ ↑	OA ↑	AJ ↑	$\langle \delta_{\text{avg}}^x \rangle$ ↑	OA ↑	AJ ↑	$\langle \delta_{\text{avg}}^x \rangle$ ↑	OA ↑
<i>Supervised</i>									
Kubric-VFS-Like [16]	51.9	69.8	84.6	33.1	48.5	79.4	40.5	59.0	80.0
COTR [22]	40.1	60.7	78.6	35.4	51.3	80.2	19.0	38.8	57.4
TAP-Net [10]	65.4	77.7	93.0	38.4	53.1	82.3	46.6	60.9	85.0
PIPs [17]	59.1	74.8	88.6	42.0	59.4	82.1	35.3	54.8	77.4
TAPIR [11]	–	–	–	56.2	70.0	86.5	49.6	64.2	85.0
CoTracker [23]	–	–	–	61.8	76.1	88.3	49.6	64.3	83.3
CoTracker3 [24]	–	–	–	63.8	76.3	90.2	55.8	68.5	88.3
LocoTrack [7]	–	–	–	62.9	75.3	87.2	52.9	66.8	85.3
BootsTAPIR [12]	–	–	–	61.4	73.6	88.7	54.6	68.4	86.5
PIPs++ [54]	–	–	–	–	73.7	–	–	63.5	–
<i>Self-supervised</i>									
CRW-C [51]	31.4	48.1	76.3	7.7	13.5	72.9	20.2	33.6	70.6
CRW-D [51]	35.8	52.4	80.9	23.6	38.0	77.2	21.9	36.8	70.4
DIFT-C [44]	28.3	45.2	69.0	18.1	33.0	68.8	19.8	33.7	68.7
DIFT-D [44]	41.6	59.8	83.9	29.7	48.2	77.2	19.5	34.4	70.1
Flow-Walk-C [3]	49.4	66.7	82.7	35.2	51.4	80.6	40.9	55.5	84.5
Flow-Walk-D [3]	51.1	68.1	80.3	24.4	40.9	76.5	46.9	65.9	81.8
ARFlow-C [32]	52.3	68.1	81.4	35.0	51.8	79.7	27.3	44.3	79.5
GMRW-C [42]	54.2	<u>72.4</u>	82.6	<u>41.8</u>	<u>60.9</u>	78.3	31.9	52.3	72.9
GMRW-D [42]	51.4	71.7	83.9	30.3	49.4	77.3	36.3	59.2	71.0
<i>Video-GMAE</i> zeroshot	<u>54.3</u>	67.0	91.9	41.3	55.7	<u>85.2</u>	<u>60.1</u>	<u>69.1</u>	<u>90.7</u>
<i>Video-GMAE</i> base frozen	60.8	70.7	96.6	45.2	55.2	91.4	61.7	68.9	97.1
<i>Video-GMAE</i> base finetune	73.6	82.3	97.5	55.7	66.1	92.1	75.0	81.7	97.7
<i>Video-GMAE</i> large frozen	62.4	71.9	96.6	46.7	55.8	91.8	65.1	72.0	97.4
<i>Video-GMAE</i> large finetune	74.0	82.4	97.6	57.9	67.7	93.5	75.1	81.6	97.9

Table 2. Performance comparison on three video datasets. *Video-GMAE* shows strong performance across multiple datasets and multiple metrics. To compare with all the models, we use a stride of five for evaluations. We bold the best supervised performance metrics and underline the best self-supervised performance metrics.

in Table 2, we compare our *Video-GMAE*-zeroshot against self-supervised approaches and *Video-GMAE*-frozen and *Video-GMAE*-finetune against the supervised approaches.

7.4. Frame Length Scaling

In this experiment, we investigate how our representations evolve as we train on longer frame sequences. We perform this ablation with *Video-GMAE*-base on frame lengths $\{2, 4, 8, 16, 24\}$. We evaluate these models on the strided TAPVid [10] evaluation protocol, once with the stride equal to the number of frames the model was trained on, and once with a stride of two for a faithful comparison across the different

ablations. We evaluate these using AJ on TAP-Vid Davis, and show results in Table 3.

We find that training on longer frame sequences and evaluating on longer stride lengths leads to reduced AJ numbers. This is likely due to longer frame lengths inducing stronger regularization from correspondence-aware pretraining, which limits the learned representations’ quality. When comparing on a fixed stride of two, we find that future information helps tracking to a certain extent. Having four and eight frames of information helps the model predict at a stride of two, but increasing it to 16 and 24 seems to decrease AJ due to the stronger regularization at these frame lengths.

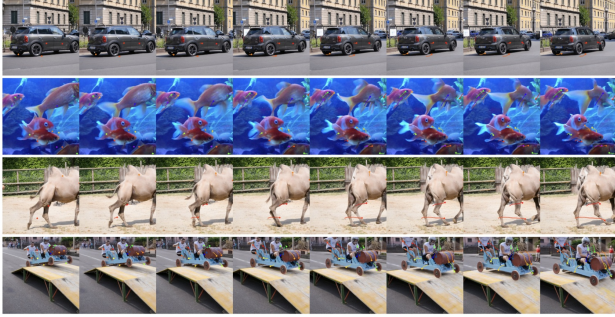


Figure 7. **Qualitative Results from Finetuned Model:** We show qualitative results (16 frames, we only show every other frame here) from our *Video-GMAE*-base encoder finetuned model, after fine-tuning on Kubric datasets [16]. Our model was able to achieve state-of-the-art performance on point tracking, and was able to track points over long range with high precision. A higher-resolution version can be found in the appendix (Fig. 11).

Delta Ablation			# Frames vs. Stride		
$\Delta\mu_{ij}$	Δr_{ij}	AJ $\uparrow$	# Frames (F)	Stride = F	Stride = 2
✓	✓	44.7	2	55.4	55.4
✓	–	44.4	4	49.2	64.2
–	✓	42.5	8	41.5	63.5
–	–	39.1	16	47.8	57.5
			24	24.7	52.4

Table 3. Ablations on finetuned tracking. Left: effect of using mean and RGB deltas during pretraining on Davis AJ. Right: impact of the number of frames and stride on Davis AJ for *Video-GMAE*-base, comparing stride equal to the number of frames (F) and a fixed stride of two.

7.5. Delta Gaussian Ablations

We also explore the importance of the delta Gaussian parameters in modeling temporal evolution. In Table 3, we ablate three variants using *Video-GMAE*-large: (1) integrating only the mean components $\Delta\mu_{ij}$, (2) integrating only the RGB components Δr_{ij} , and (3) integrating neither of them, and having static Gaussians. This helps isolate the contribution of motion versus appearance changes in temporal modeling. By evaluating AJ on TAP-Vid Davis, we find that integrating mean correspondence not only enables zeroshot point tracking but also improves the learned encoder latents for fine-tuned point tracking.

7.6. Tracking Similar-looking Objects

To probe tracking under appearance ambiguity, we construct a Kubric-based benchmark where each scene contains 2-5 near-identical instances of the same object category, making identity difficult to resolve from appearance alone. We render and evaluate on 100 scenes at 256×256 with MOVi-style assets and HDRI backgrounds. As shown in Fig. 8, both *Video-GMAE*-zeroshot and GMRW-C qualitatively struggle

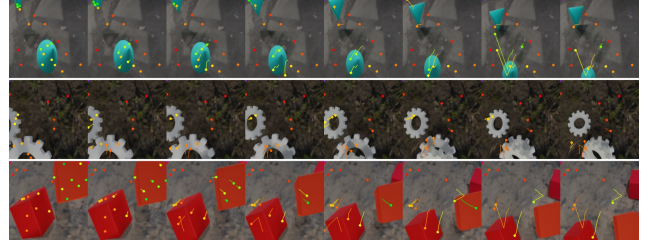


Figure 8. Qualitative examples of our designed Kubric-based benchmark probing identity tracking under near-identical appearances.

in this setting, but their quantitative results are comparable: *Video-GMAE*-zeroshot achieves 51.1 AJ, versus 48.1 for GMRW-C.

8. Limitations

One of the main limitations of our work is the modeling of the camera. Throughout the pretraining, we assume a static camera, but this is not an ideal assumption when pretraining on internet-style videos. Because of the assumption about the static camera, we also fail to recover any metric 3-D information in the 3-D Gaussians. Another assumption we made in this work is regarding the correspondence-based regularization. While this is a weaker assumption than the prior one, and works reasonably well on short videos, when the number of frames at pretraining becomes very large, this regularization starts to hurt the learning. Additionally, during the pretraining, we only use 256 Gaussians per frame, which significantly limits the quality of our renderings. These limitations also restrict the quality of the Gaussian representation; addressing them should yield even better zeroshot tracking performance.

9. Societal Impact

Our method reduces reliance on labeled tracking data by learning correspondence from raw videos, which may broaden access to video representation learning for applications such as robotics and scientific video analysis. At the same time, improved tracking capabilities can be misused in surveillance or other privacy-sensitive settings. Although our work does not address identity recognition and is developed in a research setting, we encourage careful deployment with appropriate consent, transparency, and oversight.

10. Conclusion

We presented *Video-GMAE*, a self-supervised video pretraining method that represents videos as 3-D Gaussians evolving over time. By predicting Gaussian deltas across frames, the model is encouraged to learn temporal correspondence rather than only patch reconstruction. This produces strong representations for tracking, enabling competitive zeroshot point tracking and strong performance after fine-tuning across multiple benchmarks. Our results suggest that structured, correspondence-aware generative objectives are a promising alternative to standard patch-based video pretraining.

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